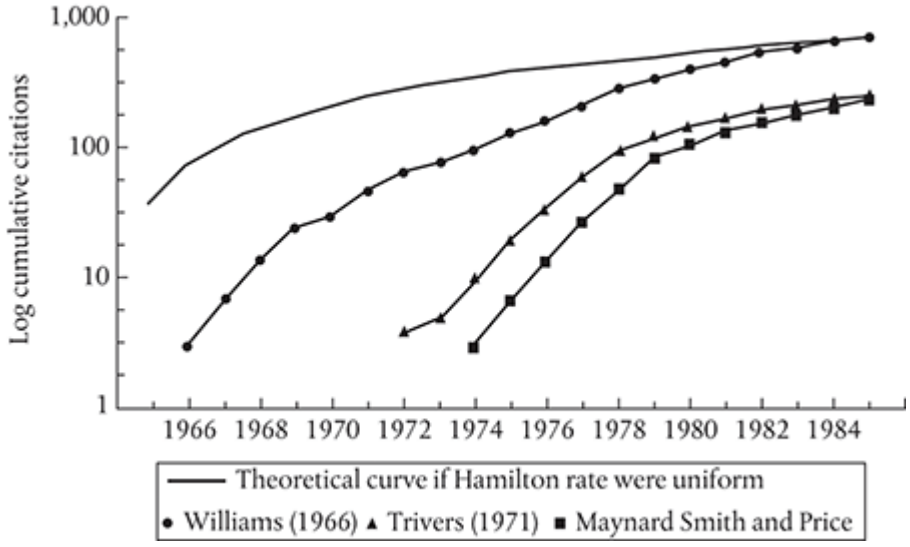


Figure 3. Log cumulative citations of three works not by Hamilton, compared with 'theoretical' curve for Hamilton (details explained in text)

Secondly, one might be tempted to think that there is something, if not inevitable, at least trivially expected about an exponential increase. Isn't the whole rate of publication of scientific papers, and therefore opportunities to cite other papers, itself increasing exponentially? Perhaps the size of the scientific community is increasing exponentially. To show that there is something special about the Hamilton meme, the easiest way is to plot the same kind of graph for some other papers. Figure 3 also shows the log cumulative citation frequencies of three other works (which incidentally were also highly influential on the first edition of this book). These are Williams's 1966 book, *Adaptation and Natural Selection*; Trivers's 1971 paper on reciprocal altruism; and Maynard Smith and Price's 1973 paper introducing the ESS idea. All three of them show curves that clearly are not exponential over the whole time-span. For these works too, however, the annual citation rates are far from uniform, and over part of their range they may even be exponential. The Williams curve, for instance, is approximately a straight line on the log scale from about 1970 onwards, suggesting that it, too entered an explosive phase of influence.

I have been downplaying the influence of particular books in spreading the Hamilton meme. Nevertheless, there is one apparently suggestive postscript to this little piece of memic analysis. As in the case of 'Auld Lang Syne' and 'Rule Britannia', we have an illuminating mutant error. The correct title of Hamilton's 1964 pair of papers was 'The genetical evolution of social behaviour'. In the mid to late 1970s, a rash of publications, *Sociobiology* and *The Selfish Gene* among them, mistakenly cited it as 'The genetical theory of social behaviour'. Jon Seger and Paul Harvey looked for the earliest occurrence of this mutant meme, thinking that it would be a neat marker, almost like a radioactive label, for tracing scientific influence. They traced it back to E. O. Wilson's influential book, *Sociobiology*, published in 1975, and even found some indirect evidence for this suggested pedigree.

Much as I admire Wilson's *tour de force*—I wish people would read it more and read about it less—my hackles have always risen at the entirely false suggestion that his book influenced mine. Yet, since my book also contained the mutant citation—the 'radioactive label'—it began to look alarmingly as though at least one meme had travelled from Wilson to me! This would not have been particularly surprising, since *Sociobiology* arrived in Britain just as I was completing *The Selfish Gene*, the very time when I would have been working on my bibliography. Wilson's massive bibliography would have seemed a godsend, saving hours in the library. My chagrin